



OutreachBey — Enterprise Docker Deployment Guide

This guide details how to seamlessly deploy and scale **OutreachBey** on a Linux Cloud VPS (DigitalOcean, Linode, AWS, or Vultr) for your \$299/mo Pro Tier clients.

Pre-Flight: Modify Your `index.js`

To make your code automatically adapt between your local Mac (headless: false, system Chrome) and your Cloud VPS (headless: true, Docker Linux Chrome, proxies), update the Puppeteer configuration in your `index.js` file:

```
const isDocker = process.env.IS_DOCKER === 'true';
const proxyUrl = process.env.PROXY_URL || '';

// Cross-Platform Executable Paths
const chromePath = isDocker
  ? '/usr/bin/google-chrome-stable'
  : '/Applications/Google Chrome.app/Contents/MacOS/Google Chrome';

// Chromium Performance Sandbox Arguments
const puppeteerArgs = [
  '--no-sandbox',
  '--disable-setuid-sandbox',
  '--disable-dev-shm-usage',
  '--disable-gpu',
  '--no-first-run',
  '--no-default-browser-check'
];

// Dynamically Route Automated Web Traffic Through Proxy
if (proxyUrl) {
  // Expected format: http://username:password@ip:port
  puppeteerArgs.push(`--proxy-server=${proxyUrl}`);
}

const client = new Client({
  authStrategy: new LocalAuth({
    dataPath: './.webjs_auth' // Mounted to the host filesystem via Docker vol
  }),
  puppeteer: {
    executablePath: chromePath,
    headless: isDocker ? true : 'new', // Runs completely headless on Cloud VPS
    args: puppeteerArgs
  }
});
```

Step-by-Step Production Host Setup

Step 1: Install Docker on your Cloud VPS

Log into your remote server via SSH and execute the official installation:

```
# Update local packages
sudo apt update && sudo apt upgrade -y

# Install Docker
curl -fsSL [https://get.docker.com](https://get.docker.com) -o get-docker.sh
sudo sh get-docker.sh

# Install Docker Compose (v2)
sudo apt install docker-compose-plugin -y
```

Step 2: Initialize Directory Structure

Run this single terminal script on your server to set up an organized, multi-tenant workspace with correct file permissions:

```
mkdir -p outreachbey/clients/normans-cay/data outreachbey/clients/butler-boats/
cd outreachbey

# Create empty fallback tracking logs
touch clients/normans-cay/followups.json clients/butler-boats/followups.json
echo "[]" > clients/normans-cay/followups.json
echo "[]" > clients/butler-boats/followups.json

# Seed the CSV data directories (Upload your clients' files here)
# Format: contacts.csv, settings.csv, log.csv must be dropped in each client fo
```

Your final folder tree on the VPS must look like this:

```
outreachbey/
├── Dockerfile
├── docker-compose.yml
├── index.js
├── package.json
├── clients/
│   ├── normans-cay/
│   │   ├── data/
│   │   │   ├── contacts.csv
│   │   │   ├── settings.csv
│   │   │   └── log.csv
│   │   └── followups.json
│   └── butler-boats/
│       ├── data/
│       │   ├── contacts.csv
│       │   ├── settings.csv
│       │   └── log.csv
```

└─ followups.json

Launching and Onboarding Clients

Step 3: Build & Spin Up Client Instances

To compile the lightweight, customized Linux Chrome container and launch all clients in background daemon mode:

```
docker compose up --build -d
```

Step 4: Scan Client QR Codes

Because the container runs headless, you must view the output logs to display the terminal QR code for device pairing. Run:

```
# Display the log stream for Norman's Cay Boutique
docker logs -f outreachbey-normans-cay
```

Have Norman open **WhatsApp** → **Settings** → **Linked Devices** → **Scan Device** on his phone and point it at the Terminal. Once successfully paired, the session will save to the host folder and run 24/7.

Container Operations & Maintenance

Common Maintenance Commands

```
# View running status of all client agents
docker compose ps

# Force restart a specific client's agent container
docker compose restart normans-cay

# Stop all client containers without destroying persistent data
docker compose down

# Re-build and pull updates after code modifications
docker compose up -d --build
```

Critical Troubleshooting

1. WhatsApp Web Session Keeps Dropping/Logging Out

- **Cause:** Puppeteer crashed out of memory or ran out of disk space, corrupting the dynamic authorization cache.
- **Fix:** Clean-restart the container. If that fails, delete the authentication directory on the host VPS and trigger a fresh QR code pairing scan:

```
docker compose stop normans-cay
sudo rm -rf clients/normans-cay/wwwbjs_auth
docker compose start normans-cay
docker logs -f outreachbey-normans-cay
```

2. Zombie Chrome Processes Eating Host RAM

- **Cause:** Container was forcefully terminated without allowing Puppeteer's internal process manager to close gracefully.
- **Fix:** Docker's restart policy `restart: unless-stopped` automatically garbage collects dead child runtimes, but if your VPS RAM is maxed out, run:

```
# Check system RAM utilization
free -m
# Clear memory cache
sudo sysctl -w vm.drop_caches=3
```